Eixo 1

Gabriel Luiz Espindola Pedro, Paulo Ricardo Branco da Silva,
Dimas Irion Alves, and Anatel authors

***Abstract*— Assessing the quality of telecommunications services across municipalities with heterogeneous structural, geographic, and socioeconomic conditions requires methodologies that go beyond simple weighted averages. This paper introduces a Graph Neural Network (GNN)-based manifold ranking framework that learns a latent representation of municipality-level telecom quality indicators and produces a unified ranking score. The proposed architecture employs Residual Graph Convolutional layers operating on a k-nearest-neighbor graph constructed from 18 infrastructure and quality-of-service indicators monitored by the Brazilian National Telecommunications Agency (ANATEL). The model is trained with a multi-objective loss function that simultaneously enforces pairwise ranking consistency, input reconstruction fidelity, geometric smoothness on the learned manifold, and directional alignment with domain knowledge. A monotonicity constraint is imposed on the scoring head to guarantee that improvements in underlying indicators always translate into higher scores. We validate the approach on a longitudinal dataset covering 4,088 Brazilian municipalities over 30 monthly periods (July 2022–December 2024), totaling over 82,000 municipality–period observations. Ablation studies across six experimental configurations demonstrate the contribution of each loss component. Temporal stability analyses via spaghetti plots reveal that the manifold ranker produces smoother and more consistent rankings than the regulatory baseline (IQS), while geographic visualizations confirm alignment with known infrastructure patterns. The framework provides a principled, data-driven alternative to expert-defined composite indices for regulatory quality assessment at national scale.**

***Keywords*— Learn to Rank (LTR), Manifold Learning, Pareto Optimization, Multi Objective Optimization, Quality of Service (QoS)**

I. Introduction

Queremos ranquear o servico de telecom dos municipios do brasil (porque?), isso é importante (porque?)

Brazilian National Telecommunications Agency (ANATEL) Regulation on the Quality of Telecommunications Services (RQUAL) established the Service Quality Index (IQS), a linear combination of Quality of Service (QoS) indicators that summarizes the quality of service provisioning on a given municipality as way to estimate the overall service quality.

When we want to compare service quality across municipalities we face a problem, most of the IQS values are grouped in a small group in the upper part of the support, making them indistinguishable one from another.

Gabriel Luiz Espindola Pedro, Paulo Ricardo Branco da Silva, and Dimas Irion Alves are with Telecommunications Department of Aeronautics Institute of Technology, São José dos Campos, SP, Brazil, e-mails: {gabrielluizep, pbranco, dimasirion}@ita.br; Anatel authors are with Anatel e-mails: {xxx, yyy}@anatel.gov.br. This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) – Finance Code 001, Anatel under Decentralized Execution Agreement (TED) No. 03/2023, Work Program: 24.722.2205.20ZD.0001, and São Paulo Research Foundation (FAPESP) under grant 2020/09838-0 (BIOS—Brazilian Institute of Data Science).

As the assumption of equality of service provisioning in multiple municipality stands improbable given the number of variable composed of, sample noise, differences of infrastructure, geographical distribution, etc., we can assume that when comparing two municipalities we always have one better than another. By repeating this process of comparison, and using the underlying transitivity property, we can define a ranking of municipalities service quality.

Given the problem, there is no such thing as "better" municipality, because the absence of labeling identifying the ordering between them we have to work with a auto-supervised model, where I find a way to understand in mid training if I'm converging to a solution that makes sense to my goal.

To enable relevant comparison we have to represent the municipalities with some fit mathematical structure. Since we don't have a definition or examples of which municipalities are better than other, and by how much, we try to bring the municipality similarity by using the IQS in a k-Nearest Neighbors (kNN) graph, where we have the edges between the municipalities $\mathcal{E}$ representing the similarity between them.

Geometrically, we can understand this process of generating the kNN $\mathcal{E}$ graph as modeling of the manifold where the data lives in.

The remainder of this work is presented as follows: Section **??** builds the intuition on the frameworks used; Section **??** [...]

motivação

o ranqueamento é difícil por questão do contexto e das diferentes dimensões de análise

o ranqueamento é útil para comparações técnicas, econômicas e diagnósticas

as técnicas de ranqueamento em problemas multi-dimensionais são difíceis porque os problemas possuem várias dimensões de análise

por que exatamente a existência de problemas multi-objetivo de otimização torna o problema de otimização mais complexo?

como Pareto entra para resolver esse tipo de questão

POr que são problemas sem rotulamento?

Explicar como os nossos custos exploram os aspectos do problema e cobrem diferentes perspectivas de otimização e contribuem a uma solução mais balanceada e mais "ótima".

Resultados de como o nosso esquema de otimização contribui para um cenário de otimização melhorado em face a

otimizações sobre variáveis escalares (otimização unidimensional).

Demonstrar que esse tipo de análise aumenta o range para variáveis alvo e como faz isso

Demonstrar ablações para demonstrar que o uso da técnica de fato gera bons resultados e que não é mera aleatoriedade.

II. Preliminaries

III. Problem setup

A. IQS

B. Graphs

IV. Dataset

The dataset is organized as a panel indexed by two keys: *month*, which identifies the observation period, and *municipality*, the official Brazilian Institute of Geography and Statistics (IBGE) numeric code that uniquely identifies each municipality. Each row therefore corresponds to one municipality–month pair. The feature columns fall into four groups. The first group contains coverage indicators: the 5G resident coverage variable reports the share of residents in the municipality covered by a 5G signal, normalized by the total resident population, while the 5G deployment indicator is a binary variable encoding whether 5G technology is available in the municipality. The second group consists of composite quality indices derived from the regulatory framework: IBC is the Index of Coverage Quality, ISG is the General Satisfaction Index, and IQS is the Quality of Service index, which aggregates the eight technical indicators into a single score. The third group comprises the eight individual QoS indicators defined under Anatel's RQUAL regulation [**?**], each normalized to $[0, 1]$ with higher values denoting superior performance: IND_1 (voice-call connection rate), IND_2 (voice-call drop rate, complemented), IND_3 (data-connection rate), IND_4 (download and upload speed compliance rate), IND_5 (data-connection latency, complemented), IND_6 (jitter, complemented), IND_7 (packet-loss rate, complemented), and IND_8 (service availability rate). The fourth group contains throughput measurements collected through Anatel's INF4 speed-testing infrastructure, reported separately by radio-access technology and direction: $\mathrm{INF4}_{\mathrm{DL,3G}}$, $\mathrm{INF4}_{\mathrm{DL,4G}}$, and $\mathrm{INF4}_{\mathrm{DL,5G}}$ record median download speeds over 3G, 4G, and 5G connections, respectively, while $\mathrm{INF4}_{\mathrm{UP,3G}}$, $\mathrm{INF4}_{\mathrm{UP,4G}}$, and $\mathrm{INF4}_{\mathrm{UP,5G}}$ record the corresponding upload speeds.

A. Legacy IQS

The Anatel legacy composite, $\mathrm{QoS}_{\mathrm{leg}}$, aggregates the K normalized technical quality-of-service indicators through a fixed regulator-defined convex combination,

$$\mathrm{QoS}_{\mathrm{leg}} = y_{\max} \sum_{k=1}^{K} w_k^{\mathrm{leg}} \, \mathrm{IND}_k, \tag{1}$$

where $y_{\max} = 10$ is the upper bound of the user satisfaction scale, each $\mathrm{IND}_k \in [0, 1]$ is a normalized technical indicator with higher values denoting superior performance, and the weight vector $w^{\mathrm{leg}} \in \Delta^{K-1}$ lies on the probability simplex $\Delta^{K-1} = \{w \in \mathbb{R}^K : w_k \geq 0, \sum_k w_k = 1\}$. Under the legacy specification, the weights are fixed at $w^{\mathrm{leg}} = (0.1, 0.1, 0.1, 0.2, 0.1, 0.1, 0.1, 0.2)$, assigning greater relevance to IND_4 (download and upload speed compliance rate) and IND_8 (service availability rate), while distributing equal weight among the remaining indicators.

B. Revised IQS

The Anatel revised composite, $\mathrm{QoS}_{\mathrm{rev}}$, introduced in 2025 through the updated Reference Values Document (DVR) [**?**], replaces the fixed weights with a performance-dependent scheme combined with a coverage bonus. The indicators are first ranked from worst-performing to best-performing within each municipality, and the weight assigned to rank position r is

$$\alpha_r = \frac{1}{r \, H_K}, \tag{2}$$

where $H_K = \sum_{j=1}^{K} 1/j$ is the K-th harmonic number. The revised composite is then defined as

$$\mathrm{QoS}_{\mathrm{rev}} = y_{\max} \sum_{r=1}^{K} \alpha_r \cdot \mathrm{IND}_{\pi(r)} + B_{\mathrm{cov}}, \tag{3}$$

where π is the permutation ordering indicators from worst to best performance, and B_{cov} is a coverage bonus corresponding to a fixed increment of five points, added whenever coverage with 4G or higher technology in the municipality's urban area reaches at least 95%. When $B_{\mathrm{cov}} = 0$, the revised rule reduces to a linear function of the indicators; when $B_{\mathrm{cov}} \neq 0$, it becomes affine. The inverse-harmonic weighting assigns greater importance to the worst-performing indicators, penalizing municipalities with more critical technical dimensions, in contrast to the static weight distribution of the legacy composite.

V. Manifold Ranker Model

A. Training Objectives and Loss Terms

1) Pairwise Ranking Loss:

2) Reconstruction Loss:

3) Geometric Smoothness Loss:

4) Total Loss and Weighting:

$$\mathcal{L} = \lambda_{\mathrm{rank}} \mathcal{L}_{\mathrm{rank}} + \lambda_{\mathrm{recon}} \mathcal{L}_{\mathrm{recon}} + \lambda_{\mathrm{geo}} \mathcal{L}_{\mathrm{geo}} + \lambda_{\mathrm{compass}} \mathcal{L}_{\mathrm{compass}} \tag{4}$$

VI. Results and Discussion

VII. Conclusions

Acknowledgments

References

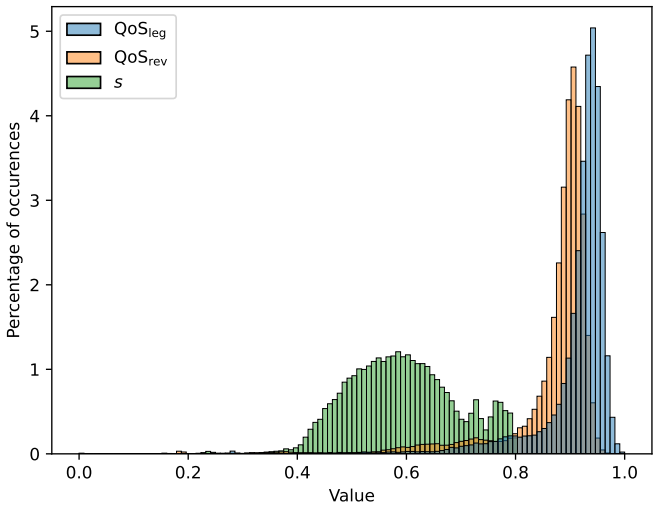
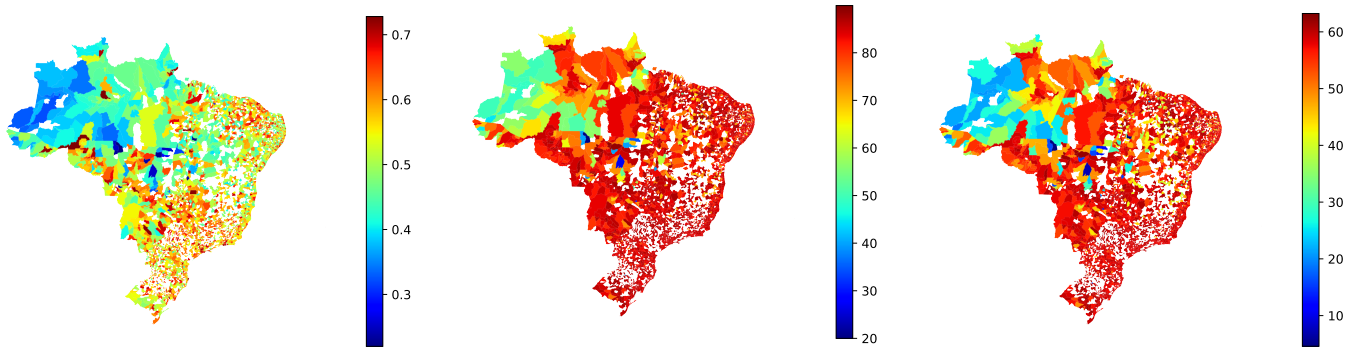


Fig. 1: Enter Caption

Rank	ID	Municipality	s
5	3106200	Belo Horizonte	0.801732
12	5103403	Cuiabá	0.800890
13	3205200	Vila Velha	0.800725
18	3303302	Niterói	0.800296
22	4314902	Porto Alegre	0.800049
24	3300456	Belford Roxo	0.799928
26	4113700	Londrina	0.799900
34	5300108	Brasília	0.799486
41	5201405	Aparecida de Goiânia	0.799363
44	2408102	Natal	0.799303
45	2927408	Salvador	0.799278
48	5108402	Várzea Grande	0.799234
49	5208707	Goiânia	0.799225
54	3118601	Contagem	0.799126
62	1302603	Manaus	0.798931
65	1500800	Ananindeua	0.798808
79	3304557	Rio de Janeiro	0.798561
80	5002704	Campo Grande	0.798516
82	2700300	Arapiraca	0.798476
99	3157807	Santa Luzia	0.798180



(a) Manifold Ranker s (b) $\mathrm{QoS}_{\mathrm{leg}}$ (c) $\mathrm{QoS}_{\mathrm{rev}}$

Fig. 2: Enter Caption